

Notes: Sharpe (JF, 1964)

Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk

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1 Big picture

- **Question.** In a capital market with risky and riskless assets, how are asset prices determined in equilibrium, and what kind of risk is actually priced?
- **Setting.** A one-period capital market in which investors choose portfolios using a mean-variance logic, can borrow or lend at a common pure interest rate, and share homogeneous expectations about returns and correlations.
- **Core challenge.** Total risk of an individual asset cannot be the relevant pricing object, because diversification can eliminate part of that risk. A positive equilibrium theory must explain which component remains relevant after investors optimize.
- **Main theoretical result.** In equilibrium, risky portfolios available to investors line up with a **capital market line**: every investor combines some efficient risky portfolio with borrowing or lending at the risk-free rate.
- **Asset-pricing result.** The expected return on an individual asset is increasing not in its **total** risk, but in its **systematic risk**, measured by how strongly its return co-moves with an efficient portfolio. In modern notation, this is the asset's **beta**.
- **Main intuition.** Unsystematic risk can be diversified away, so competition in capital markets only rewards the part of risk that survives diversification.
- **Why this paper matters.** This paper is the classic equilibrium bridge from Markowitz-Tobin portfolio theory to what later became the CAPM tradition. It turns a normative portfolio-choice argument into a positive theory of capital asset prices.

2 Data and measurement (key objects)

This is a **theory paper**; there is no empirical dataset. The relevant “objects” are portfolio returns, risk measures, beliefs, and equilibrium prices.

2.1 Economic units and one-period structure

- **Investor.** An investor allocates a given amount of current wealth, denoted W_I , across risky assets and a riskless asset.
- **Terminal wealth.** Future wealth is W_t .
- **Rate of return.** The return on the investor's portfolio is

$$R = \frac{W_t - W_I}{W_I}.$$

- **Investment plans.** A plan may consist of a single asset or a portfolio of assets. Each plan is represented by its expected return and its standard deviation.

2.2 Beliefs and preference objects

- **Expected future wealth and risk.** Investors evaluate a plan using

$$E_w \quad \text{and} \quad \sigma_w,$$

the expected value and standard deviation of terminal wealth.

- **Equivalent return representation.** Because terminal wealth is directly linked to the portfolio return, the same preferences can be written in terms of

$$E_R \quad \text{and} \quad \sigma_R.$$

- **Correlation coefficients.** The co-movement of investment plans matters through correlations such as r_{ab} for plans a and b , or r_{ig} for asset i and efficient portfolio g .

2.3 Key equilibrium objects

- **Pure rate of interest P :** the expected return on the riskless asset.
- **Investment opportunity curve:** the lower right-hand boundary of feasible risky portfolios in (E_R, σ_R) space.
- **Capital market line:** the straight line obtained when the riskless asset is combined with an efficient risky portfolio.
- **Efficient risky combination ϕ or g :** a risky portfolio that lies on the capital market line.
- **Systematic-risk coefficient B_{ig} :** Sharpe's notation for the asset's responsiveness to an efficient portfolio. In modern language, this is a beta-like object.

2.4 Unobservables and timing

- **Subjective expectations.** The expected returns, standard deviations, and correlations are belief objects.
- **Homogeneous expectations.** The equilibrium analysis assumes that all investors agree on these belief objects.
- **Static timing.** The model is one-period: investors choose today, returns are realized once, and there is no multi-period dynamic optimization.

2.5 Conceptual applications

- The model explains why **defensive** securities should earn lower expected returns than **aggressive** securities.
- It separates the **price of time** (the pure interest rate) from the **price of risk** (the excess expected return per unit of systematic risk).
- It explains why diversification changes the relevant notion of risk for valuation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Investor preference function

Sharpe assumes that an investor evaluates any investment using only the mean and standard deviation of terminal wealth:

$$U = f(E_w, \sigma_w). \quad (1)$$

Because terminal wealth is a linear function of the portfolio return, the utility function can be written as

$$U = g(E_R, \sigma_R). \quad (2)$$

- **Interpretation.** The investor prefers higher expected return and dislikes higher risk.
- **Economic content.** This is a mean-variance representation of portfolio choice. It is the backbone of the paper's equilibrium analysis.

3.2 Feasible plans, efficiency, and the investment opportunity curve

Every feasible investment plan can be represented by a point in the (E_R, σ_R) plane. A plan is **efficient** if there is no alternative with the same expected return and lower risk, the same risk and higher expected return, or both higher expected return and lower risk.

- **Interpretation.** The investor problem can be split into two steps: first identify the efficient set, then choose the preferred point on that set.
- **Geometry.** The efficient set is the lower right-hand boundary of the feasible risky region — the **investment opportunity curve**.

3.3 Combining two risky investment plans

If a fraction α of wealth is invested in risky plan A and the remaining share $(1 - \alpha)$ in risky plan B , the expected return of the combination is

$$E_{Rc} = \alpha E_{Ra} + (1 - \alpha) E_{Rb}. \quad (3)$$

The standard deviation is

$$\sigma_{Rc} = \sqrt{\alpha^2 \sigma_{Ra}^2 + (1 - \alpha)^2 \sigma_{Rb}^2 + 2r_{ab}\alpha(1 - \alpha)\sigma_{Ra}\sigma_{Rb}}. \quad (4)$$

- **Interpretation of r_{ab} .** This is the correlation between the returns of the two risky plans.
- **Diversification result.** If $r_{ab} < 1$, the combination risk lies below the straight line between A and B in mean-standard-deviation space. This is the basic diversification gain.
- **Implication.** Total asset risk by itself is not informative about price, because the relevant risk of an asset depends on what it is combined with.

3.4 Introducing the riskless asset and the capital market line

Let P denote a riskless asset with expected return equal to the pure interest rate and zero risk. Combining it with a risky asset A yields

$$E_{Rc} = \alpha E_{Rp} + (1 - \alpha) E_{Ra}, \quad (5)$$

and

$$\sigma_{Rc} = \sqrt{\alpha^2 \sigma_{Rp}^2 + (1 - \alpha)^2 \sigma_{Ra}^2 + 2r_{pa}\alpha(1 - \alpha)\sigma_{Rp}\sigma_{Ra}}. \quad (6)$$

Because $\sigma_{Rp} = 0$, this simplifies to

$$\sigma_{Rc} = (1 - \alpha)\sigma_{Ra}. \quad (7)$$

Hence any combination of the riskless asset and a risky portfolio must lie on a **straight line** in (E_R, σ_R) space.

- **Interpretation.** The efficient frontier with a riskless asset becomes a straight line from the riskless point to the tangent risky portfolio.
- **Two-fund separation.** Investors first choose an optimal risky combination ϕ and then borrow or lend at the pure rate to reach the desired point on the capital market line.
- **Important nuance.** Even if borrowing is impossible or limited, the tangent portfolio still plays the central role for many investors.

3.5 Capital-market equilibrium

Sharpe imposes two strong assumptions:

- all investors can borrow or lend at a **common pure interest rate**;
- all investors have **homogeneous expectations** about returns, risks, and correlations.

Under these assumptions, every investor evaluates the opportunity set in the same way. If some set of prices makes one efficient risky combination ϕ especially attractive, all investors want portfolios built around that same risky combination and then differ only in how much they borrow or lend.

Price adjustment then works as follows:

- prices of assets in ϕ rise, reducing their expected returns;
- prices of assets not in ϕ fall, increasing their expected returns;
- the opportunity set rotates until every asset enters at least one efficient risky combination.

In equilibrium, the efficient boundary becomes a straight line. Sharpe effectively writes the capital market line as

$$\sigma_R = s(E_R - P), \quad (8)$$

where P is the pure interest rate and s is the slope coefficient.

- **Interpretation.** The market offers two prices: the price of time, P , and the price of risk, reflected in the slope of the capital market line.

- **Subtle point.** The paper does *not* say that every investor must literally hold the same risky portfolio. Rather, many efficient risky combinations can coexist, but they all lie on the same line and must therefore be perfectly positively correlated.

3.6 Pricing an individual asset: the tangency argument

Take any individual risky asset i and any efficient risky combination g that contains it. Consider all feasible combinations of asset i and portfolio g . Their return locus must be tangent to the capital market line at g .

For a combination with share α in asset i and $(1 - \alpha)$ in combination g , the risk of the combination is

$$\sigma = \sqrt{\alpha^2 \sigma_{Ri}^2 + (1 - \alpha)^2 \sigma_{Rg}^2 + 2r_{ig}\alpha(1 - \alpha)\sigma_{Ri}\sigma_{Rg}}, \quad (9)$$

while expected return is

$$E = \alpha E_{Ri} + (1 - \alpha) E_{Rg}. \quad (10)$$

Tangency between this locus and the capital market line implies

$$\frac{\sigma_{Rg} - r_{ig}\sigma_{Ri}}{E_{Rg} - E_{Ri}} = \frac{\sigma_{Rg}}{E_{Rg} - P}. \quad (11)$$

Now define Sharpe's systematic-risk coefficient

$$B_{ig} = \frac{r_{ig}\sigma_{Ri}}{\sigma_{Rg}}. \quad (12)$$

Then the expected return on asset i satisfies

$$E_{Ri} = P + B_{ig}(E_{Rg} - P). \quad (13)$$

- **Interpretation.** This is the paper's central pricing formula. Expected return depends on an asset's responsiveness to an efficient portfolio, not on its own total standard deviation.
- **Modern interpretation.** Equation (13) is the core CAPM logic. In modern notation, B_{ig} is a beta and g is typically interpreted as a market-like efficient portfolio.
- **Economic meaning.** An asset earns a higher expected return only if it contributes nondiversifiable risk to an efficient portfolio.

3.7 Systematic versus unsystematic risk

Sharpe interprets B_{ig} using a regression-style decomposition of the asset return R_i on the return of efficient combination R_g .

- The slope B_{ig} measures the predicted response of R_i to movements in R_g .
- The portion of risk coming from co-movement with R_g is **systematic risk**.
- The remaining part, uncorrelated with R_g , is **unsystematic risk**.
- **Interpretation.** Unsystematic risk is not priced because it can be diversified away.
- **Pricing lesson.** The cross section of expected returns is linear in systematic-risk exposure.

3.8 Why any efficient combination can serve as the benchmark

Sharpe's final step is to show that one can choose *any* efficient combination as the benchmark portfolio in equation (13). Because all efficient combinations lie on the same capital market line, they are perfectly positively correlated.

- **Interpretation.** The relevant benchmark is not tied to a single arbitrary portfolio; any efficient portfolio delivers the same pricing relation.
- **Broader interpretation.** Sharpe suggests that the common factor behind efficient portfolios may be the general level of economic activity. If so, what matters for valuation is the asset's responsiveness to the economy.

4 Identification and instruments (what makes the estimates credible)

Because this is a theory paper, there are no econometric instruments. The relevant issue is what assumptions are doing the identifying work in the equilibrium argument.

4.1 What fails in the naive approach

- A naive asset-pricing rule would relate expected return to the **total** risk of an asset.
- Sharpe shows this cannot be right because investors diversify, and diversification removes the unsystematic component of total risk.
- Therefore, a pricing rule based only on σ_{Ri} is not theoretically coherent in equilibrium.

4.2 What identifies the capital market line

- Investors choose portfolios using a mean-variance ranking.
- They face a common riskless borrowing/lending rate.
- They share homogeneous expectations.
- Market prices adjust until all assets are included in some efficient risky combination.

These assumptions force the equilibrium opportunity set into a linear capital market line.

4.3 What identifies the pricing of individual assets

- The pricing equation for an individual asset does not come from a regression run on data; it comes from the **tangency condition** between the locus of combinations of asset i with an efficient portfolio g and the capital market line.
- This tangency converts a geometric statement into a pricing equation.
- The key identifying object is the asset's covariance structure with the efficient portfolio, summarized by B_{ig} .

4.4 Why systematic risk alone is priced

- If a part of an asset's risk is orthogonal to efficient portfolios, investors can diversify it away.
- Since that component does not survive in efficient aggregate holdings, competitive equilibrium gives it no separate reward.
- The only risk that remains relevant is the part that moves with the common efficient benchmark.

4.5 What is restrictive in the theory

- **Homogeneous expectations** are strong and likely unrealistic.
- **Equal borrowing and lending rates** are also strong, especially in real-world credit markets.
- The theory is static and one-period, so it abstracts from intertemporal hedging motives and richer dynamic risks.
- The paper is explicit that the assumptions are restrictive; its defense is that the implications are economically powerful and close to classical financial doctrine.

4.6 Relation to the literature

- Markowitz provided the normative portfolio-selection framework.
- Tobin showed that, with a riskless asset, portfolio choice can be broken into choice of a risky fund plus a lending/borrowing decision.
- Sharpe's contribution is to push this logic into **market equilibrium** and derive an asset-pricing relation for individual securities.
- This is why the paper is foundational: it is the positive asset-pricing counterpart to the earlier normative portfolio theory.

5 Tables and figures

The paper has **nine figures and no tables**. The figures are the whole empirical content of the paper in the sense that they summarize the geometric logic of the theory.

5.1 Figure 1: The capital market line as the classroom benchmark

Read:

- The figure shows the familiar linear trade-off between expected return and risk.
- The horizontal intercept is the **pure interest rate**, which Sharpe calls the price of time.
- The slope captures the **price of risk**, or extra expected return per unit of risk.
- The paper begins by saying that this picture is widely used but lacked a rigorous equilibrium foundation; the rest of the paper provides that foundation.

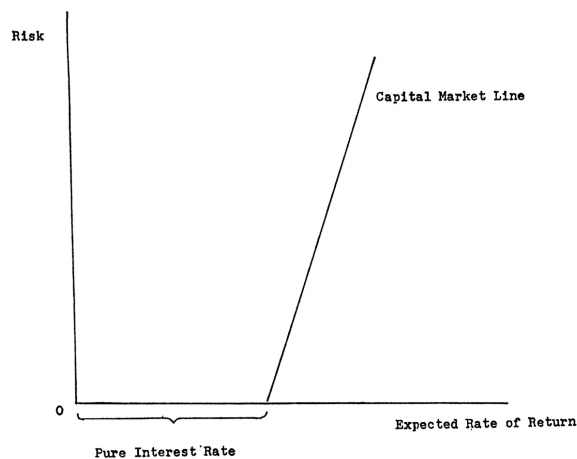


FIGURE 1

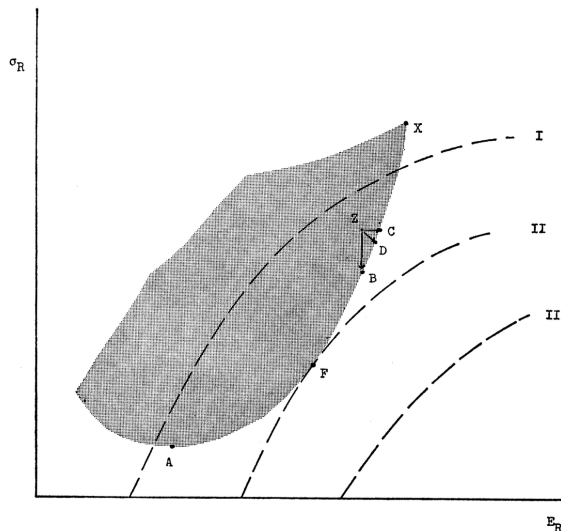


FIGURE 2

5.2 Figure 2: The risky feasible set and the efficient boundary

Read:

- The shaded region contains feasible risky portfolios in (E_R, σ_R) space.
- The lower right-hand boundary is the **investment opportunity curve**.
- An investor picks the highest attainable indifference curve, here at point F .
- The figure shows why the investor problem naturally splits into finding the efficient set first and then choosing a point on it.

5.3 Figure 3: Diversification between two risky plans

Read:

- If two risky plans are perfectly positively correlated, combinations lie on the straight line between them.
- If their correlation is lower, the combination set bends inward.
- This inward bend is the gain from diversification.
- The figure is the visual reason total risk alone cannot determine price: the same asset's contribution to portfolio risk depends on what it is combined with.

5.4 Figure 4: Adding the riskless asset

Read:

- Once a riskless asset is available, feasible portfolios lie on straight lines connecting the riskless point P to risky portfolios.

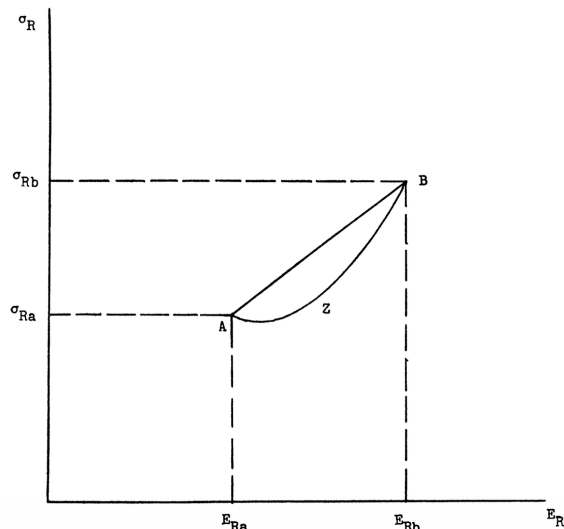


FIGURE 3

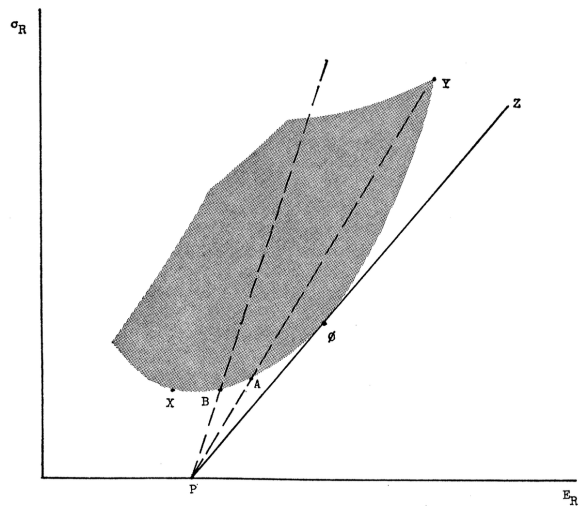


FIGURE 4

- The dominant line is the one tangent to the risky frontier at ϕ .
- Borrowing extends that same line beyond ϕ .
- This is the key geometric step behind separation: the risky portfolio choice is separated from the investor's risk tolerance.

5.5 Figure 5: Prices away from equilibrium

Read:

- At some nonequilibrium price vector, all investors want some version of the same risky combination ϕ .
- Conservative investors lend and hold ϕ ; intermediate investors hold ϕ only; aggressive investors borrow to lever up ϕ .
- Assets not in ϕ are under-demanded, while assets in ϕ are over-demanded.
- This figure explains the direction of price adjustment that moves the market toward equilibrium.

5.6 Figure 6: Equilibrium in the capital market

Read:

- In equilibrium, the capital market line becomes the efficient boundary.
- Every asset must appear in at least one efficient risky combination that lies on this line.
- Several risky combinations may be efficient at once; the theory does not require a unique risky portfolio for every investor.
- The crucial implication is that all such efficient combinations are perfectly positively correlated.

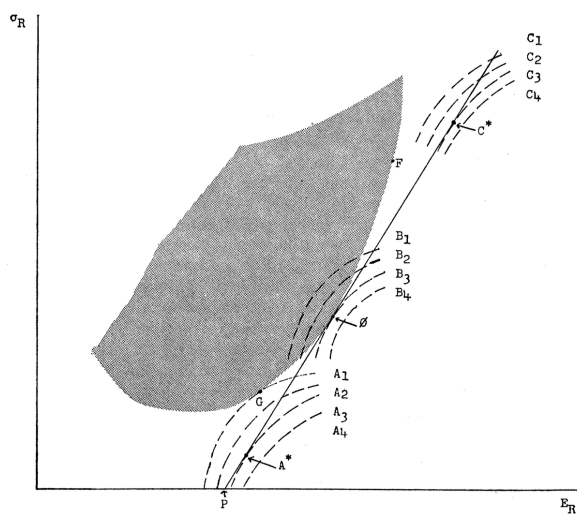


FIGURE 5

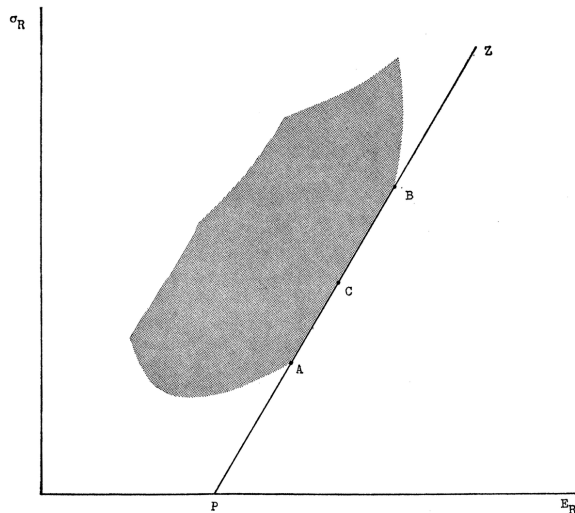


FIGURE 6

IV. THE PRICES OF CAPITAL ASSETS

5.7 Figure 7: Asset i and efficient combination g

Read:

- The figure studies all feasible mixtures of a single asset i and an efficient portfolio g .
- The locus through i , g , and g' must be tangent to the capital market line at g .
- That tangency is what generates the pricing equation for asset i .
- So Figure 7 is the bridge from portfolio geometry to security pricing.

5.8 Figure 8: Regression interpretation of beta

Read:

- The scatter plot interprets the return on asset i as linearly related to the return on efficient combination g .
- The slope B_{ig} is the asset's systematic-risk coefficient.
- Variation around the regression line is unsystematic risk.
- This figure translates the geometric theory into the modern covariance/beta language.

5.9 Figure 9: Expected return and systematic risk

Read:

- The vertical axis is B_{ig} and the horizontal axis is the expected return of asset i .
- Assets with zero systematic-risk exposure earn the pure interest rate.
- Assets with higher B_{ig} must offer higher expected return.

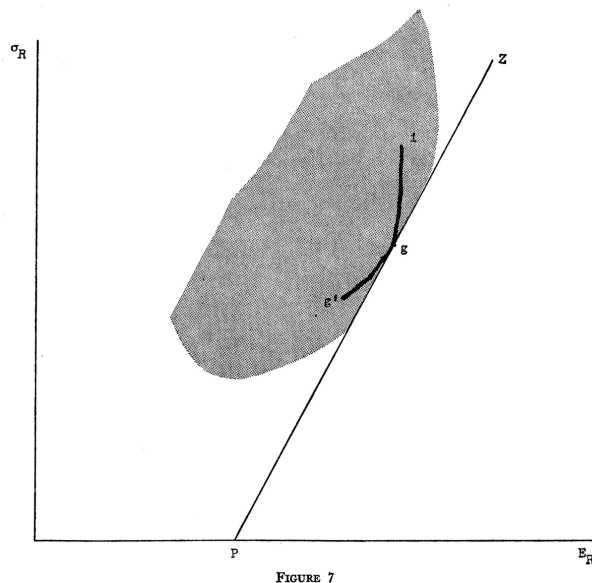


Figure 1: Enter Caption

- This is the final asset-pricing statement of the paper: there is a linear relation between expected return and nondiversifiable risk.

6 Short summary

- **Conclusion.** Sharpe develops a positive equilibrium theory of asset prices under risk using mean-variance portfolio logic plus market clearing.
- **Core insight.** The capital market line explains the pricing of efficient portfolios, while the tangency argument explains the pricing of individual assets.
- **Main lesson.** Expected return is determined by **systematic risk** alone; unsystematic risk is not priced because it can be diversified away.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper gives an equilibrium derivation of the capital market line.
- It shows that investors choose among risky assets in a way that can be separated from their borrowing/lending decision once a common riskless rate is available.
- It then shows that the expected return on any individual capital asset is linear in its systematic-risk exposure to an efficient portfolio.

7.2 Economic intuition

- Diversification removes asset-specific noise, so markets do not compensate investors for bearing that noise.
- The only risk that survives efficient diversification is the part tied to common portfolio movements — or, more economically, to overall economic activity.
- Asset prices therefore adjust until higher expected return is paid only to securities that load more heavily on this common source of risk.

7.3 Takeaway

Sharpe's paper is fundamental because it answers two questions at once. At the portfolio level, it explains why equilibrium in risky and riskless assets yields a capital market line. At the individual-security level, it explains why the relevant notion of risk is beta rather than total volatility. In that sense, the paper gives the conceptual core of the CAPM: the market rewards only nondiversifiable risk, and the reward for that risk is linear.